

Matrix inequalities and norms

Marcus's inequality for the permanent of block permanents

Difficulty: extreme **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: The all-block-count inequality is a longstanding permanent conjecture; its immediate payoff is the specialized theory of PSD block permanents.

For a square matrix C of order r , write

$$\text{per } C = \sum_{\sigma \in S_r} \prod_{i=1}^r c_{i, \sigma(i)}.$$

Let $m, k \geq 2$ be arbitrary integers and let $A \in \mathbb{C}^{mk \times mk}$ be Hermitian positive semidefinite. Partition A into an $m \times m$ array of $k \times k$ blocks A_{ij} , and form

$$G = (g_{ij})_{i,j=1}^m, \quad g_{ij} = \text{per}(A_{ij}).$$

Must

$$\text{per } A \geq \text{per } G$$

always hold?

Relevance

This asks how a costly matrix polynomial behaves under block aggregation by the same polynomial. It is a precise constraint on hierarchical permanent computation for PSD matrices; positivity is assumed for the whole matrix, not separately for off-diagonal blocks.

References

- I. M. Wanless, *Lieb's permenental dominance conjecture* (2022), Conjecture 3 and the following paragraph ([primary manuscript](#)).
- F. Zhang, *An update on a few permanent conjectures*, *Special Matrices* 4 (2016), 305–316, discussion of Marcus's permanent-of-permanents conjecture ([primary manuscript](#); [journal](#)).
- E. H. Lieb, *Proofs of some Conjectures on Permanents*, *Journal of Mathematics and Mechanics* 16 (1966), 127–134, the two-block inequality ([publisher's first page](#)).

Status check — 2026-09-10

Wanless records this inequality as open and identifies Lieb's proof for $m = 2$. Searches for "Marcus conjecture", "permanent of permanents", "block permanents", "proof", and 2025–2026 found no general resolution. This entry is specifically the inequality component: some historical formulations additionally conjecture an equality characterization. The trivial block sizes $m = 1$ and $k = 1$ are omitted, and no unqualified equality characterization is imported. Lieb's general permenental dominance conjecture would imply this assertion, but this is a separately stated historical conjecture.

Audit update (2026-09-10): Rechecked Wanless Conjecture 3 and its following paragraph. The $m = 2$ theorem and a real-matrix regime are partial cases; later searches for the permanent-of-permanents inequality found no unrestricted resolution. This is a bounded literature check, not a proof that no solution exists.